User Manual

PathHunter® Detection Kit

For Use with PathHunter Cell Lines
or
Cell Lines Generated with ProLabel®/ProLink™ Vectors

For Assay Kits:

93-0001M:	2-Plate Kit
93-0001:	10-Plate Kit
93-0001L:	100-Plate Kit
93-0001XL:	500-Plate Kit
93-0001-HTS400:	1000-Plate Kit
93-0001-HTS800:	2000-Plate Kit
93-0001-HTS1M:	2500-Plate Kit

Document Number 70-142 Revision 10

Please read the entire User Manual before proceeding with the assay.

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Overview

The PathHunter® Detection Kit is to be used with PathHunter Cell Lines or cell lines made with ProLabel®/ProLink™ expression or cloning vectors. The easy-to-use assay has been successfully run in both 96-well and 384-well microplate formats. The resulting chemiluminescent signal can be read with any standard plate reader.

Assay Principle

PathHunter products utilize the Enzyme Fragment Complementation (EFC) technology for studying many modes of actions (MOAs), including reporter signaling, protein- protein interactions, receptor- dimerization, protein translocation, receptor trafficking, and receptor internalization, all of which involve the complementation of two β-galactosidase (β-gal) fragments. The larger β-gal fragment is called Enzyme Acceptor (EA) while the smaller fragment is termed Enzyme Donor (ED; also known as ProLabel or ProLink). The two fragments are inactive when apart, but the two parts complement with one another to form a functional enzyme that hydrolyzes the substrate included in the PathHunter Detection kit to generate a chemiluminescent signal that is reflective of the level of active enzyme present. The detection reagents do not contain either the EA or ED enzyme fragments of β-gal, and therefore must be used in conjunction with PathHunter cell lines expressing both a ProLabel/ProLink tagged receptor and an EA-tagged partner protein.

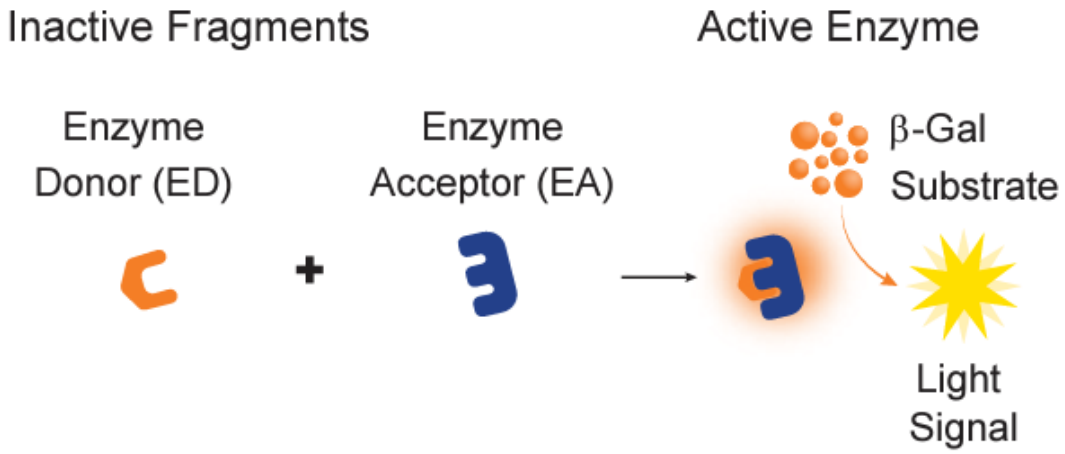

Figure 1. Enzyme Fragment Complementation (EFC) technology

Materials Provided

Catalog Number	93-0001M	93-0001	93-0001L	93-0001XL	93-0001-HTS400	93-0001-HTS800	93-0001-HTS1M
Number of Plates*	2	10	100	500	1000	2000	2500
96-well, No. of data points	~200	~1000	n/a	n/a	n/a	n/a	n/a
384-well, No. of data points	~800	~4,000	~40,000	~200,000	~400,000	~800,000	~1,000,000
Kit Components	**Total Volume (mL)**						
Cell Assay Buffer	11.4	57	418	1,825	3,650	7,300	9,125
Substrate Reagent 1	3.0	15	110	480	960	1,920	2,400
Substrate Reagent 2	0.6	3	22	96	192	384	480
Total Kit Volume	**15**	**75**	**550**	**2,401**	**4,802**	**9,604**	**12,005**

***Note**- Plates not provided with the kit.

Table 1. Materials Provided.

Storage Conditions

Upon receipt, store at -20°C. Once thawed, the reagents can be stored at 2-8°C for up to 4 weeks. For longer storage (up to the expiration date indicated on the kit box outer label), the reagents should be aliquoted in opaque containers and stored at -20°C. Once aliquoted do not freeze-thaw more than three times. Make aliquots suitable for testing one assay plate. Ensure that the reagent is equilibrated to room temperature before use in the assay for best performance.

Additional Materials Needed

Required Materials
PathHunter cell line expressing EA and ED
15 mL Polypropylene tube (or larger, if necessary)
Multimode or luminescence plate reader
Disposable reagent reservoir (Thermo Scientific, Cat. No. 8094 or similar)

Table 2. Additional Materials Needed

Assay Detection Protocol

The following procedure details the assay detection protocol to be used with PathHunter cell lines. Please refer to the user manual specific to the cell line for details about the assay preparation. Detection reagents must be prepared as a working solution prior to use. Once prepared, the working solution is stable at room temperature for up to 24 hours with no impact on assay performance.

Working Detection Solution for 96- or 384-well Format		
Components	**Volume Ratio**	**Volume per Plate (mL)**
Cell Assay Buffer	19	5.7
Substrate Reagent 1	5	1.5
Substrate Reagent 2	1	0.3
Total Volume		**7.5**

Table 3. Working Detection Solution

Volume per Plate (mL) in the table above is recommended for a single plate run. For multi-plate runs, use volume ratios stated in the table.

1. Prepare a stock of Working Detection Solution (Table 3) in a 15 mL polypropylene tube or reagent reservoir by mixing 19 parts of cell assay buffer, 5 parts of Substrate Reagent 1, and 1 part of Substrate Reagent 2.

Make stock within 24 hours of use.

2. Add 55 µL (for 96-well format) or 12.5 µL (for 384-well format) [equivalent to 50% of the assay volume] of Working Detection Solution to all wells of the assay plate.

Do not pipette up and down in the vial to mix or vortex plates.

3. Incubate assay plate for 1 hour at room temperature in the dark.

   Read samples on a standard luminescence plate reader at 0.1 to 1 second/ well for photomultiplier tube readers or 5-10 seconds for imager. The actual signal characteristics over time are affected by lab conditions such as temperature. The user should establish an optimal read time accordingly. Luminescence readout collects signal from all wavelengths. Some instrument manufacturers may include a cutoff filter at high wavelengths, but usually no wavelength setting is needed for luminescence readout

Working Detection Solution is light sensitive, thus incubation in the dark is necessary.

After addition of reagents, the samples should be read within 2 hours.

4. Data analysis can be performed using your choice of statistical analysis software (e.g., GraphPad Prism, Molecular Devices Softmax Pro, Biotek Instruments Gen5, Microsoft Excel, etc.).

Supplemental Information

Instrument Compatibility Chart	
Compatible with any luminometer. Select examples indicated below.	
Berthold Technologies: Mithras LB940, CentroLIApc	**Molecular Devices:** FLIPR, SpectraMax M3/ M4/M5/M5e, FlexStation 3, SpectraMax L
Biotek: Synergy 2	**Perkin Elmer:** TopCount, Victor 2 or V, Fusion, LumiCount, Envision, Micro-beta (Trilux), Viewlux, Northstar, EnSpire
BMG: PheraStar, Cytostar, LumiStar	**Promega:** GloMax systems
Caliper: LabChip 3000 & EZ Reader	**Tecan:** Ultra, Evolution
GE: LEAD seeker, Farcyte	**Thermo Scientific:** Luminoskan Ascent
Hamamatsu: FDS6000, FDSS/RayCatcher	**Turner BioSystems:** Modulus Microplate

Table 4. Instrument Compatibility Chart

Document Revision History

Revision Number	Date Released	Revision Details
10	June 2024	Document Revision [updated Eurofins DiscoverX format, added several high-volume kits] This document replaces the following User Manual: PathHunter® Detection Kit (70-142 Rev. 9)

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